■ AI Task Board

AI-Powered Smart Meeting Task Manager

Project Report

Full-Stack Web Application · Node.js · MongoDB · Vanilla JS

AI Task Board is a full-stack web application that converts raw meeting transcripts into structured, actionable tasks. It uses a rule-based NLP engine to detect action items, identify the person responsible, assign a priority level, and persist everything in MongoDB — giving each team member a personalised task board without any manual effort.

1. What It Does

Meetings produce dozens of action items, but those items are buried inside paragraphs of conversation. AI Task Board solves this by letting a user paste any meeting transcript — even several pages of unstructured dialogue — and automatically producing a clean task board split into three views: **All Tasks**, **My Tasks**, and **Done**.

Each extracted task carries a **priority badge** (High / Medium / Low), an **assignee tag** showing who the task belongs to, and action buttons for marking it done, editing the text, deleting it, or starring it as important. A live search bar filters all three columns simultaneously, and a stats strip at the top always shows the current task counts.

The backend is built with **Node.js + Express** and stores every user account and task state in **MongoDB**. Tasks survive page refreshes and logouts — the board picks up exactly where the user left off.

- **User Authentication:** Register and login with hashed passwords; each user gets their own isolated board.
- **Transcript Analysis:** Paste any length of raw meeting text; the parser scans for action words and names.
- **Task Extraction:** Only genuine action items are extracted — greetings, fillers and casual talk are ignored.
- **Priority Classification:** Tasks are auto-classified as High (fix/bug/error), Medium (review/check), or Low.
- **Assignee Detection:** The parser identifies who each task belongs to by reading names in the transcript.
- **My Tasks Filter:** Enter your meeting identity to instantly filter tasks assigned only to you.
- **CRUD Operations:** Mark done, edit text, delete, or star any task — all synced to MongoDB in real time.
- **Search:** Live search filters tasks by text or assignee across all three columns at once.
- **Reset Board:** One-click reset clears the board and wipes task state from the database.
- **Data Persistence:** Task arrays, done lists, and transcript history are stored per-user in MongoDB.

2. How It Works — Step by Step

Step 1 — User Registration

A new user opens the app and is presented with the registration screen. They fill in their full name, email address, password, and role (e.g. Developer, Tester). Clicking **Register** sends the details to the Express backend, which hashes the password with bcrypt and stores the new document in the *users* collection in MongoDB. The user is then redirected to the login screen.

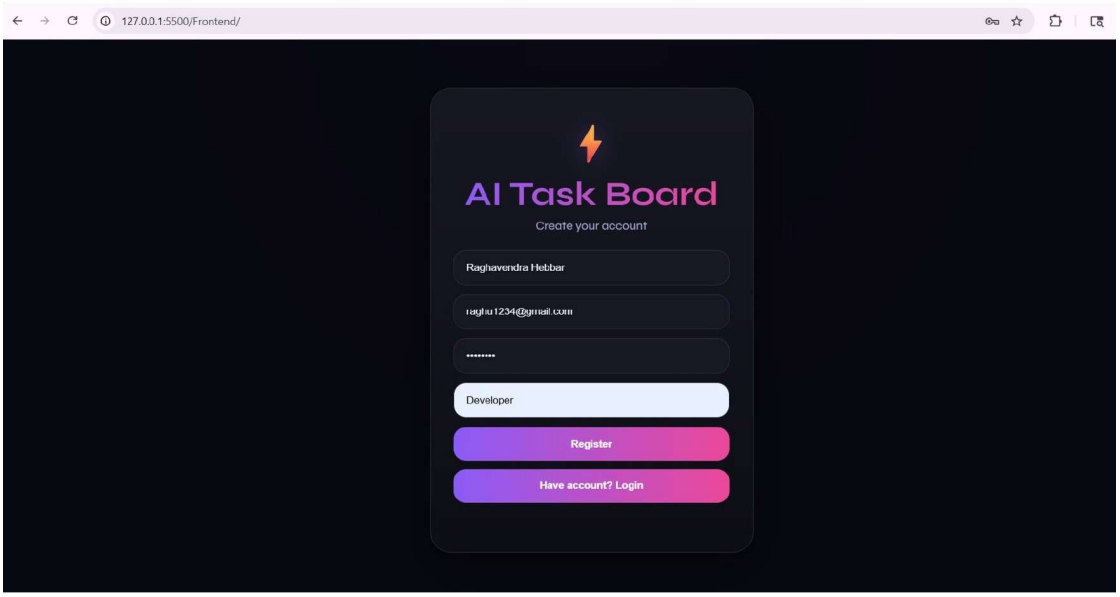

Figure 1 — Registration screen: user enters name, email, password and role

Step 2 — Login

The login screen accepts email and password. The backend validates the credentials, compares the hashed password, and returns a success response. If the user does not exist, the app shows a red *User not found* message. On success, the app transitions to the main dashboard and displays a **Welcome** toast in the bottom-right corner.

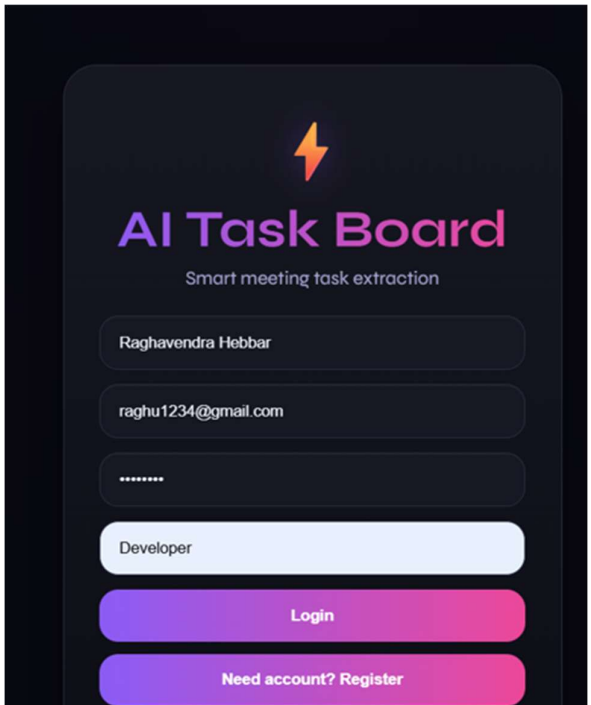

Figure 2 — Login screen

Step 3 — Main Dashboard (Empty State)

After login the user lands on the main dashboard. The top-bar shows the user's name and a live search bar. The transcript text-area and meeting-identity input are empty. The three task columns — **All Tasks**, **My Tasks**, and **Done** — show zero counts and placeholder messages. The board is now ready to receive a meeting transcript.

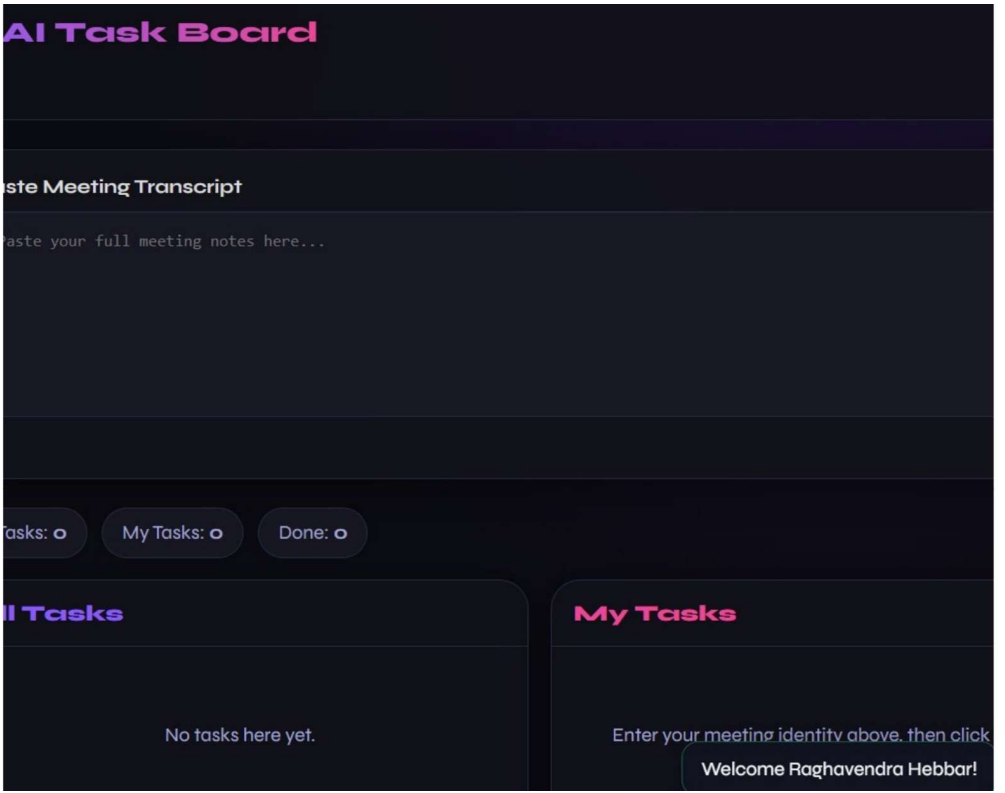

Figure 3 — Empty dashboard after login; task counts are all zero

Step 4 — Paste Meeting Transcript & Enter Identity

The user pastes the full meeting transcript into the textarea — this can be several pages of unstructured conversation. In the right panel they type their meeting identity (e.g. *Arjun*), which is the name as it appears in the transcript. This tells the app which tasks to show in the **My Tasks** column.

The user then clicks **Analyze Transcript**.

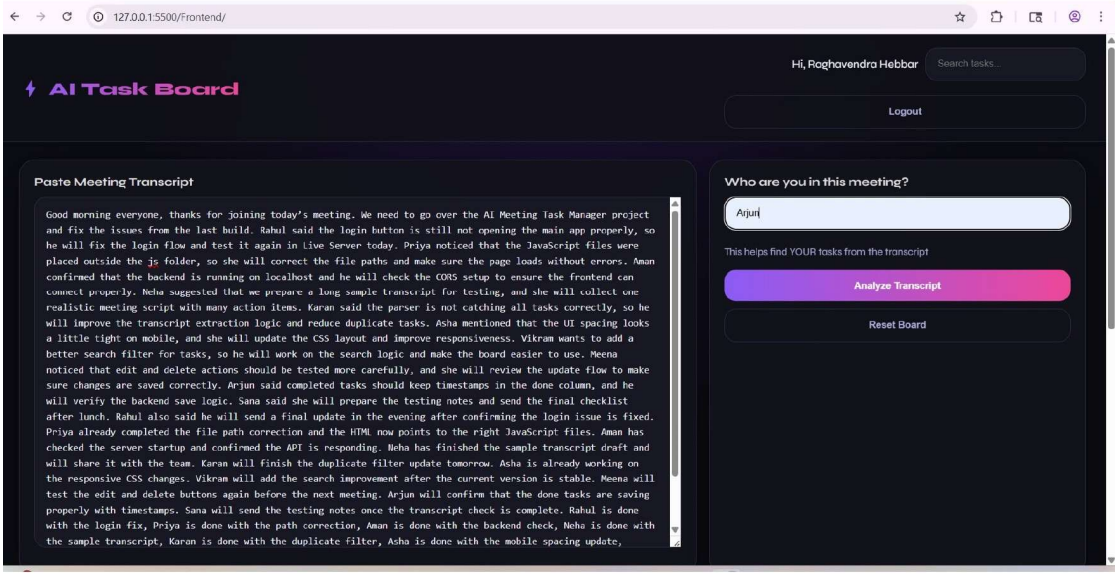

Figure 4 — Transcript pasted and meeting identity set to "Arjun" before clicking Analyze

Step 5 — Task Board After Analysis

The parser scans the transcript sentence by sentence, looks for action verbs (fix, review, test, send, update…), detects capitalised names near those verbs, splits compound sentences into individual tasks, and classifies each task's priority. The resulting tasks are rendered as cards in the **All Tasks** column. The **My Tasks** column shows only cards where the assignee matches the meeting identity. The **Done** column shows any tasks the parser detected were already completed in the transcript.

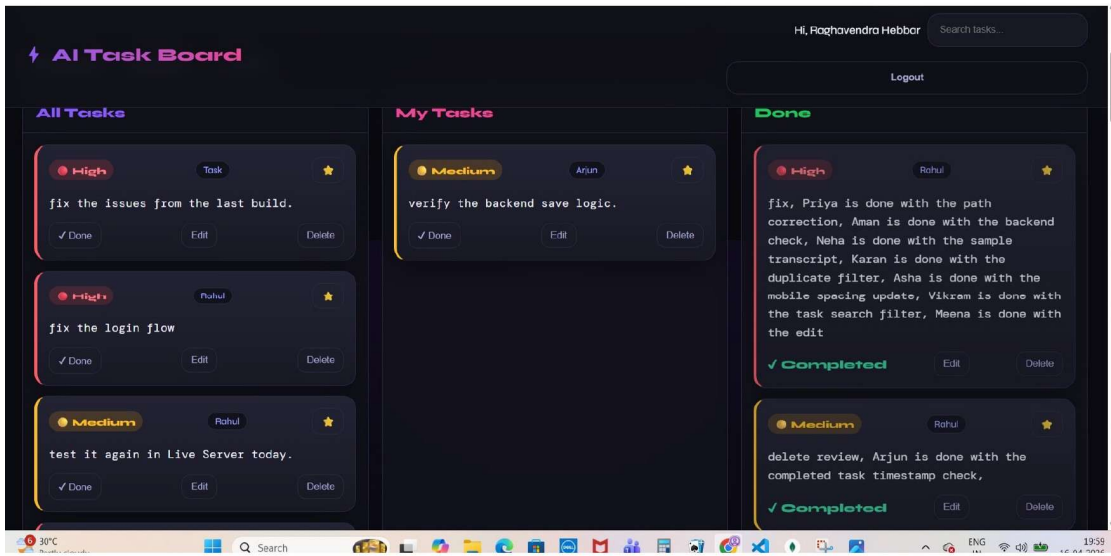

Figure 5 — Board after analysis: All Tasks (left), My Tasks for Arjun (centre), auto-detected Done tasks (right)

Step 6 — Delete a Task

Clicking the **Delete** button on any task card triggers a browser confirmation dialog. The stats bar at the top shows 25 All Tasks, 1 My Task, and 3 Done before the deletion. On confirming, the card is removed from the UI and a DELETE request is sent to the backend, which removes that task from the user's array in MongoDB.

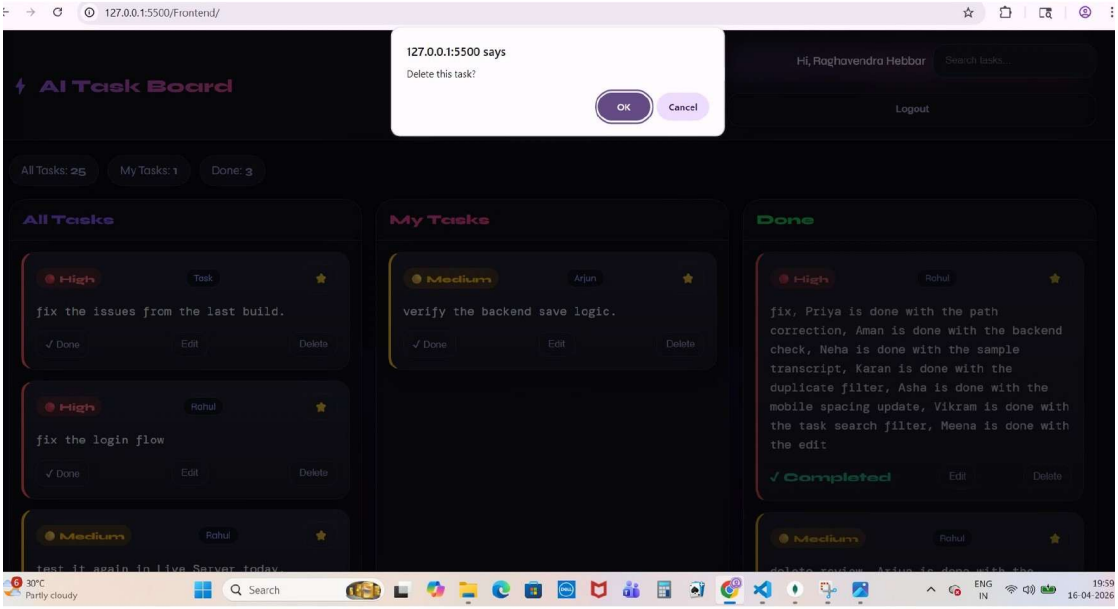

Figure 6 — Delete confirmation dialog before removing a task from the board

Step 7 — Task Deleted Confirmation

After confirming the deletion the task card disappears, the All Tasks count drops from 25 to 24, and a **Task deleted.** toast notification appears at the bottom of the screen. All other columns remain unchanged, demonstrating that only the target task was affected.

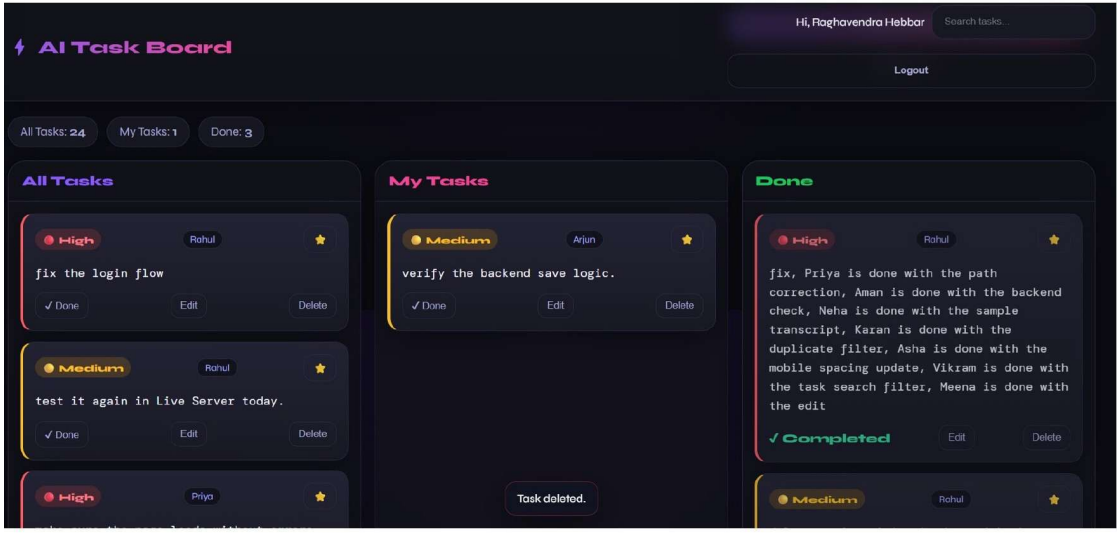

Figure 7 — Board after deletion with "Task deleted." toast; count updated to 24

Step 8 — Mark a Task as Done

Clicking the ✓ **Done** button on any active task moves it from the All Tasks or My Tasks column to the Done column. The card reappears in Done with a green ✓ **Completed** label. Simultaneously a **Task marked done.** toast confirms the action, and the backend appends the task to the user's *doneTasks* array in MongoDB with a timestamp.

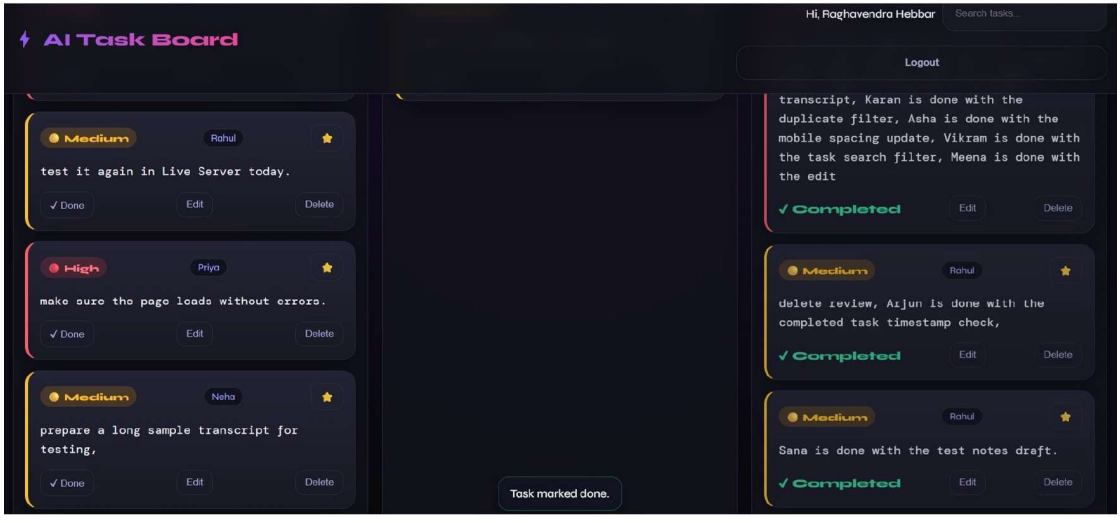

Figure 8 — "Task marked done." toast; card moves to Done column with ✓ Completed label

Step 9 — Live Search

Typing in the search bar at the top instantly filters all three columns. Only cards whose task text or assignee name contains the search string are shown. Here the user has searched for a specific phrase, and the My Tasks column highlights the matching card while unrelated cards are hidden.
Clearing the search restores the full board.

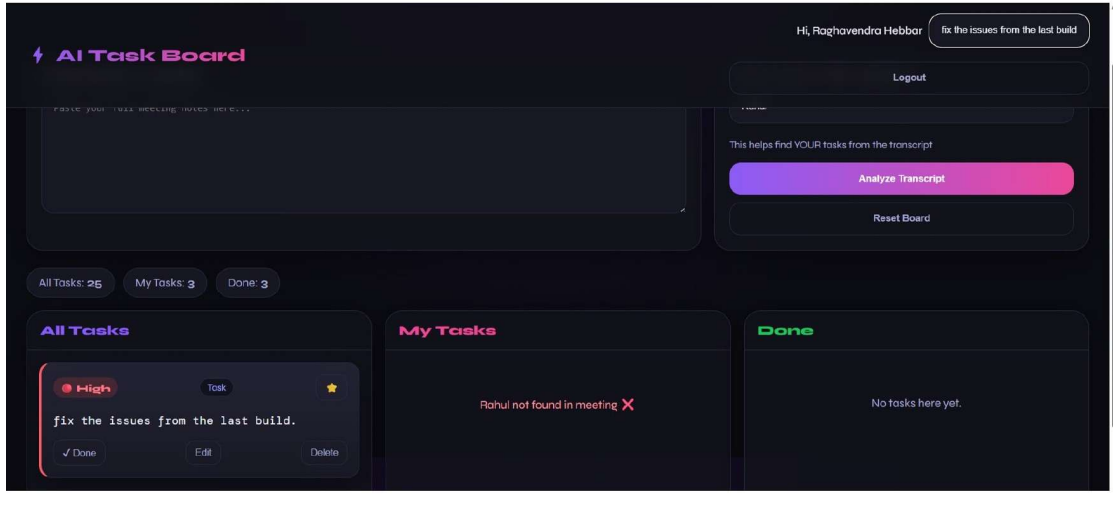

Figure 9 — Live search filtering tasks; "Rahul not found" shown when name absent from current results

Step 10 — Full Populated Board

With identity set to *Rahul* the My Tasks column shows all tasks assigned to Rahul — high-priority bugs, medium tests, and a final-update task. Each card shows its priority badge, assignee tag, and star button. The Done column on the right contains completed tasks with green ✓ **Completed** labels. The board gives a complete, at-a-glance picture of the meeting's action items.

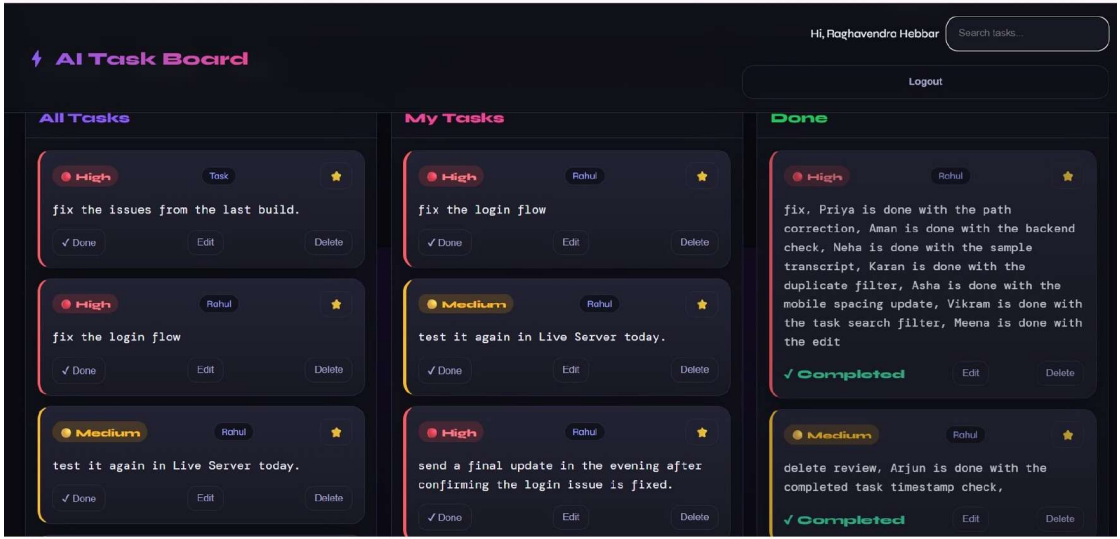

Figure 10 — Full board with All Tasks, My Tasks (Rahul), and Done columns all populated

Step 11 — Reset Board

Clicking **Reset Board** triggers a confirmation dialog asking *Reset all meeting tasks?*. On confirming, the frontend clears its in-memory arrays, wipes the transcript and identity fields, sends a POST request to the backend's */reset* endpoint, and re-renders all three columns as empty. This is useful when starting a new meeting session.

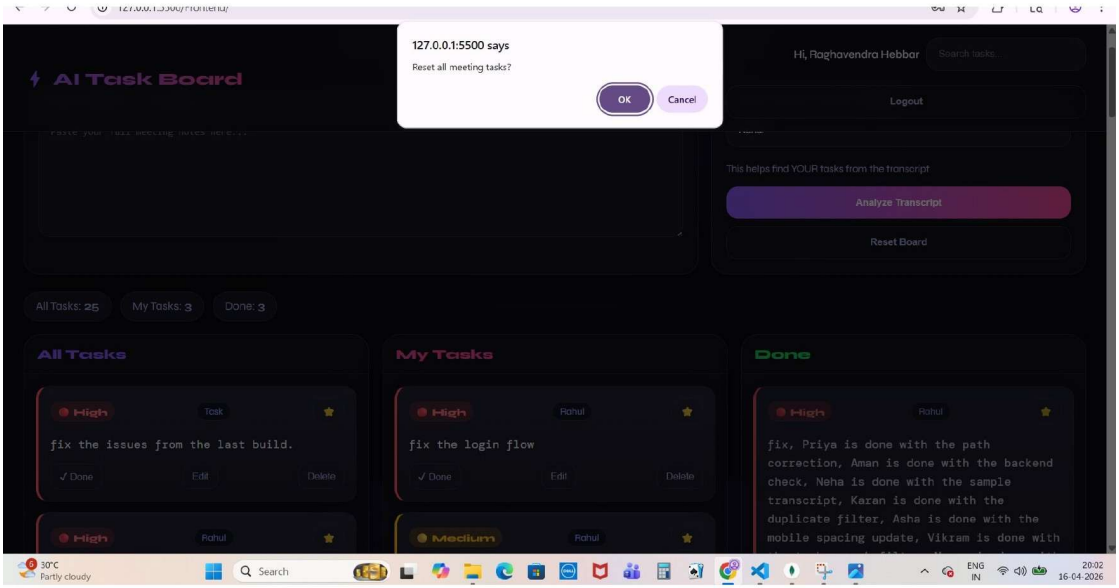

Figure 11 — Reset confirmation dialog; board shows 25 tasks before reset

Step 12 — Board After Reset

After reset all task counts return to zero and the columns display empty-state messages. A **Board reset.** toast confirms the action. The My Tasks column retains the last identity setting and shows *Rahul not found in meeting* because the board is now empty — confirming that the identity is remembered but no tasks exist yet.

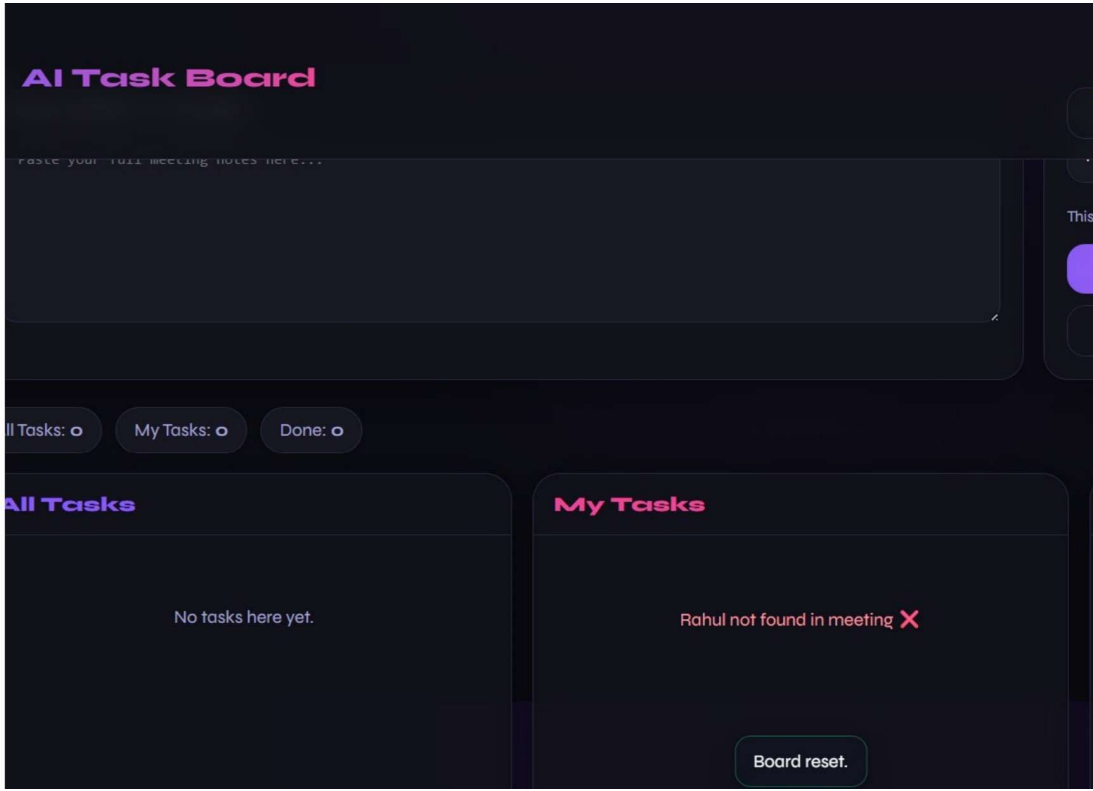

Figure 12 — Board after reset: all counts zero, "Board reset." toast visible

Step 13 — Database: Users Collection

MongoDB Compass shows the *users* collection inside the *ai_task_board* database. Each document stores the user's name, hashed email address, bcrypt-hashed password, role, and timestamps. Four registered users are visible — Jyothi R Hebbar, Geethanjali, and Raghavendra Hebbar among them. Passwords are never stored in plaintext; the *passwordHash* field contains the bcrypt output.

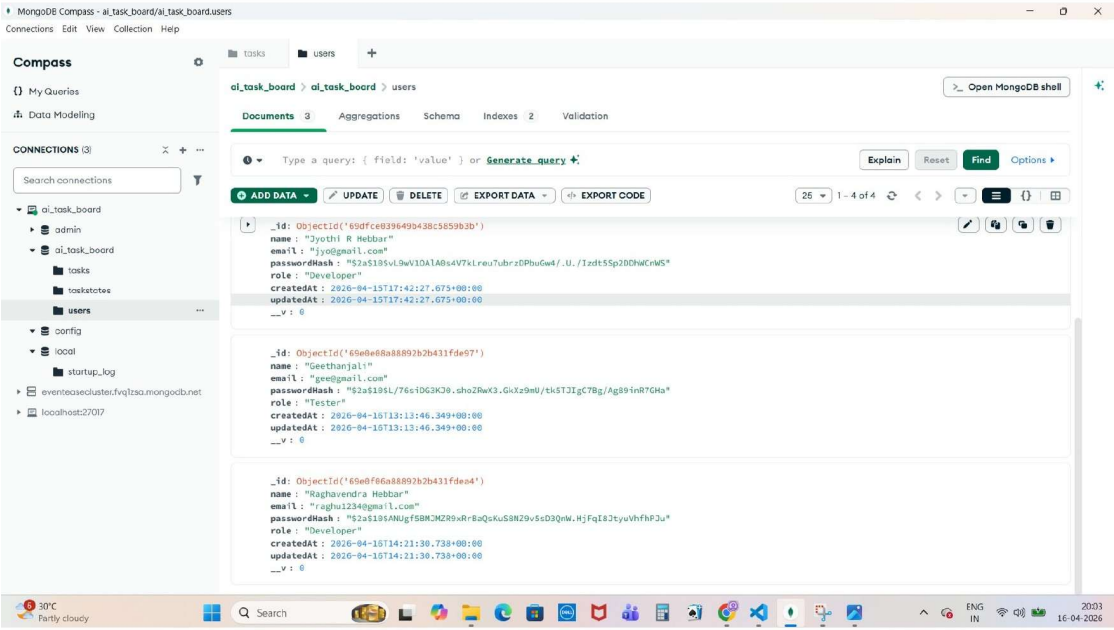

Figure 13 — MongoDB users collection showing registered users with hashed passwords

Step 14 — Database: Task State Collections

The *taskstates* collection holds the full board state per user. Each document is linked to a user via *userId* and contains three arrays: *transcriptHistory* (previous transcripts), *tasks* (active tasks), and *doneTasks* (completed tasks). The screenshot shows one user with 25 active tasks and 4 done tasks, and another with 31 tasks and 16 done tasks — proving real data is being stored and updated correctly.

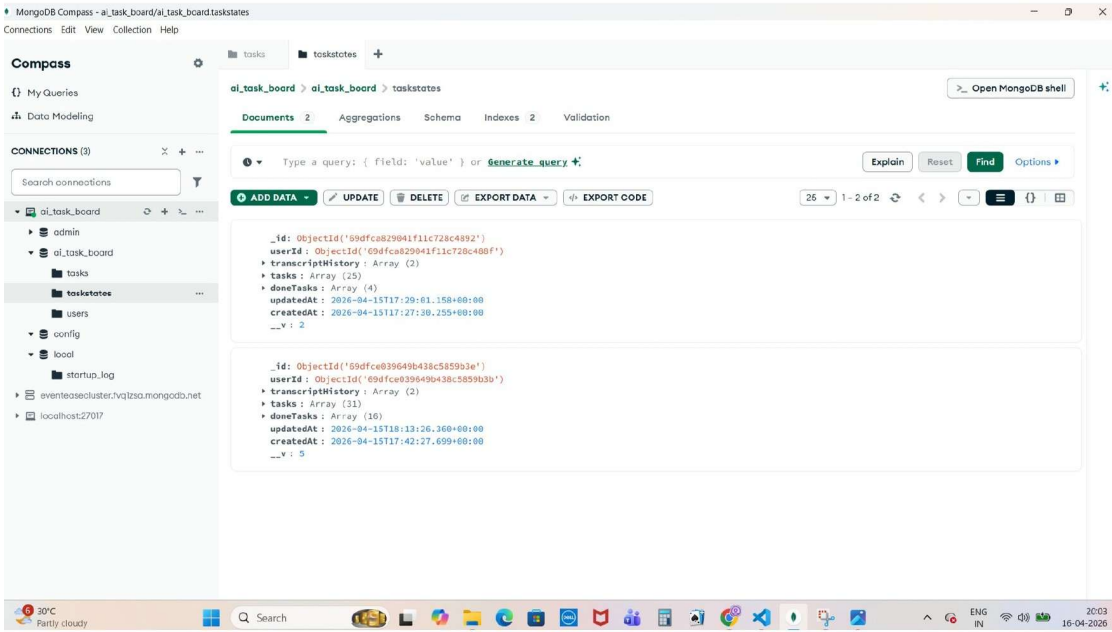

Figure 14 — MongoDB taskstates collection with tasks and doneTasks arrays per user

Step 15 — Database: Tasks Collection (Live State)

The *tasks* collection holds the current live board state. The first document shows 18 active tasks and 1 done task for one user, while two other documents from newer registrations still have empty arrays. The *updatedAt* and *createdAt* timestamps confirm when each document was last modified, giving a complete audit trail of task activity.

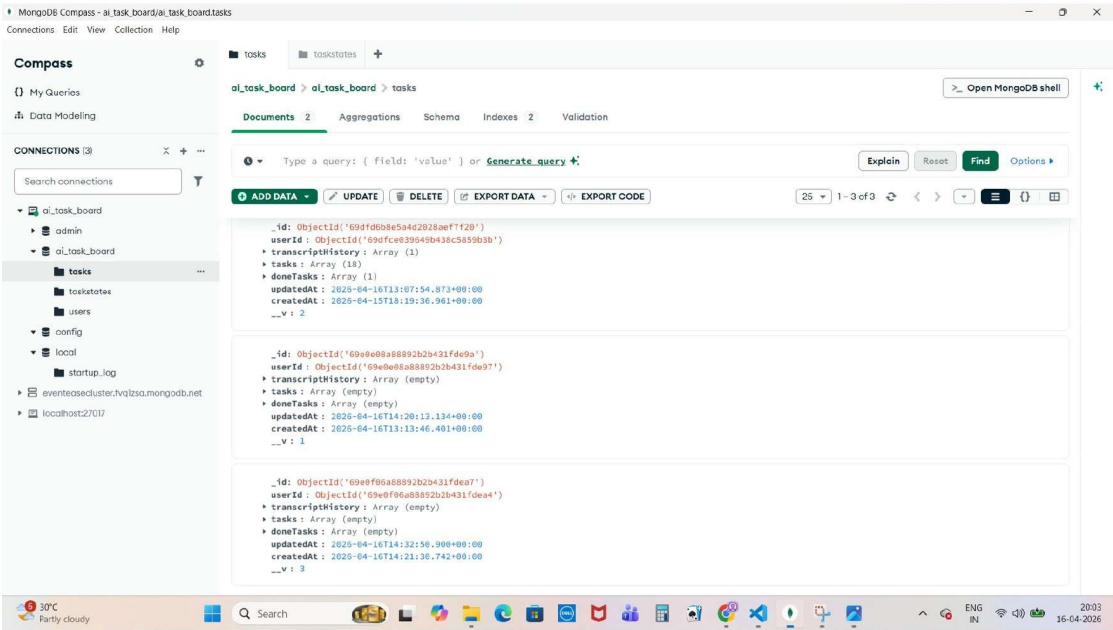

Figure 15 — MongoDB tasks collection with live task arrays and timestamps

3. Technology Stack

- **Frontend:** Vanilla HTML, CSS (dark glassmorphism UI), and JavaScript. No framework — pure DOM manipulation with the Fetch API for all backend calls.
- **Backend:** Node.js and Express.js providing a REST API with endpoints for login, register, task save/load, done tracking, and board reset.
- **Database:** MongoDB with Mongoose ODM. Three collections: users, tasks, and taskstates. Passwords hashed with bcrypt.
- **Parser Engine:** A custom rule-based JavaScript parser. Detects 40+ action verbs, extracts assignee names, splits compound sentences, classifies priority, and removes duplicates.
- **Fonts / Icons:** Google Fonts (Syne + DM Mono) for the premium typographic look. Emoji icons for priority badges and action states.
- **Development Tools:** VS Code Live Server for the frontend, nodemon for the backend during development.

4. Conclusion

AI Task Board successfully bridges the gap between raw meeting conversations and structured task management. By combining a rule-based NLP parser with a MongoDB-backed REST API and a premium dark-themed UI, the project delivers a fully functional productivity tool that can handle real-world meeting transcripts.

The system demonstrates practical skills across the full stack: frontend DOM manipulation and responsive design, backend API design with Express, database modelling and persistence with MongoDB, and intelligent text processing with a custom parser. All major CRUD operations are implemented, data is persisted per user, and the interface provides clear feedback through toast notifications and live count updates.

The project can be extended further with stronger NLP using machine learning models, real-time collaboration via WebSockets, email/push notifications for assigned tasks, an admin dashboard for team managers, and deployment to cloud platforms such as Render or Vercel for public access.